Where sample-based Krylov diagonalisation works: a localisation boundary on exactly solvable models

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Sample-based quantum diagonalisation methods (SQD, and its Krylov variant SKQD) project a Hamiltonian into the span of computational-basis configurations sampled from quantum-prepared states, betting that a modest number of sampled configurations captures the ground state. We map where that bet pays on exactly solvable spin and fermion models, where every quantity can be graded against exact diagonalisation. The governing parameter is ground-state localisation. On half-filled charge sectors of an XXZ chain (sector dimension up to 3 432) the sampled-subspace fraction required for 1% ground-energy accuracy *falls* from 84% to 2.7% of the sector as the Ising anisotropy Δ/J grows from 0.38 to 5 — a monotone boundary confirmed across 20 sampling seeds with bootstrap confidence intervals. The same boundary reappears on a 2×3 Fermi–Hubbard lattice as a function of U/t: convergence is never reached in the metallic regime ($U/t \leq 4$), while the Mott regime needs only 12.5% of the sector at $U/t = 16$. Geometry acts through the same axis: added connectivity worsens sampling only where hopping already dominates, and is irrelevant once the ground state is localised. Separately, we observe that the method's premise (a concentrated ground state) and its success (sampling *finding* that concentration) can fail independently: on the delocalised side, the fraction of the sector needed *rises* with system size even as the ground state's weight concentrates. Sampling the boundary on a superconducting processor (262 two-qubit-gate circuits, pre-registered predictions) shows the ordering survives device noise, but the localised point's subspace requirement inflates $24\times$ — attributed by a seeded control to charge-*conserving* errors polluting the frequency ranking, the corruption class that symmetry-based configuration recovery cannot see. These boundaries are consistent with why sample-based methods succeed in quantum chemistry, where a dominant reference configuration places the problem on the localised side of the axis.

1 Introduction

Sample-based quantum diagonalisation (SQD) [1] and its Krylov-inspired variants (SKQD) [2] occupy a pragmatic middle ground in near-term quantum simulation: the quantum processor is used only to *sample computational-basis configurations* from prepared states, and a classical solver diagonalises the Hamiltonian projected onto the span of the sampled configurations. The approach tolerates hardware noise gracefully — a corrupted sample is just a possibly-irrelevant basis state — and has produced striking demonstrations in quantum chemistry.

Its central bet, however, is rarely tested where it can be graded exactly: that a tractable number of sampled configurations spans the ground state. This paper maps where that bet pays, on models small enough that every claim is checked against exact diagonalisation, yet structured enough (conserved charge, tunable localisation, tunable geometry, and a fermionic analogue) that the answer has a mechanism rather than an anecdote.

Our results are simply stated. The governing parameter is *ground-state localisation* in the computational basis. Tune it — by Ising anisotropy Δ/J on a spin chain, or by interaction strength U/t on a Fermi–Hubbard lattice — and the sampled-subspace fraction required for fixed accuracy sweeps from "essentially the whole sector" to "a few percent". Geometry (1D versus 2D at identical Hilbert-space dimension) matters only on the delocalised side. And on that delocalised side the required *fraction* of the sector grows with system size even though the ground state's weight concentrates — the premise of the method and its success are separable, and the gap between them is exactly the ranking quality of the sampler.

We state the scope plainly: sectors up to dimension 3 432, one spin family and one fermion family, sampling simulated from exact statevectors (supplemented by one real-hardware configuration-recovery experiment), and single-seed measurements everywhere except the central Δ/J sweep, which carries 20 seeds and bootstrap confidence intervals. Section 6 lists the additional experiments we consider necessary before these boundaries should be treated as more than a well-controlled small-scale map.

2 Protocol

The spin family is an open XXZ chain, $H = \sum_i [J(X_i X_{i+1} + Y_i Y_{i+1}) + \Delta Z_i Z_{i+1}] + \sum_i h_i Z_i$, $J = 0.65$, fixed local fields, with conserved charge $Q = \sum_i Z_i$; all experiments run inside the half-filled sector, of dimension $\binom{n}{n/2}$: 252 ($n = 10$), 924 ($n = 12$), 3 432 ($n = 14$). The sector Hamiltonian is validated against the full 2^n matrix before anything is measured (elementwise agreement 0.0; ground energies identical to 8 decimals at $n = 4, 6$).

The protocol is SQD's actual one rather than a strawman: prepare a reference configuration (the Néel state, or the lowest-diagonal configuration of H — the spin-model analogue of a Hartree–Fock reference), evolve it to a grid of times, sample configurations from the time-evolved states (3 000 draws per time, ten times), rank configurations by observed frequency, and diagonalise H restricted to the span of the top-M. The reported quantity is the smallest M (as a fraction of the sector) whose ground energy is within 1% of the exact sector ground energy — relative to $|E_0|$ for the spin models, and relative to the sector *bandwidth* for Hubbard, where $E_0 \to 0$ at large U/t makes a relative-to-$|E_0|$ threshold measure the metric rather than the method (a bug we hit and record).

The fermionic family is a 2×3 Fermi–Hubbard lattice at half filling ($N_\uparrow = N_\downarrow = 3$, 12 qubits, sector dimension 400), mapped by an explicitly validated Jordan–Wigner construction: $[H, N_\uparrow] = [H, N_\downarrow] = 0$ exactly; at $U = 0$ the full many-body spectrum matches the free-fermion prediction to 1.9×10^{-15}; at $t = 0$ every eigenvalue is an exact integer multiple of U.

3 The localisation boundary

Spin chain. At the hopping-dominated point $\Delta/J = 0.38$ the $n = 12$ sector requires 84.4% [81.2, 86.6]% of its 924 configurations for 1% accuracy; at $\Delta/J = 1$, 48.7% [45.5, 53.0]%; at $\Delta/J = 2$, 18.9% [17.9, 20.0]%; and by $\Delta/J \geq 5$ the requirement collapses to 2.7% with zero variance across all 20 seeds (every seed converges at the smallest subspace probed). The decrease is monotone across the sweep (verified programmatically, not by eye). Fig. 1(a) shows both this requirement and the underlying ground-state concentration.

Fermi–Hubbard. The same axis, relabelled U/t [Fig. 1(b)]: in the metallic regime ($U/t = 0.5$, 2, 4) the target is *never* reached — not even in the span of every configuration the sampler observed (278, 398, 395 of 400 respectively) — while the Mott regime needs 37.5% at $U/t = 8$ and 12.5% at $U/t = 16$, where 90% of the ground state lives on 11% and 4% of the sector. This is, we believe, the practically relevant statement of why sample-based diagonalisation succeeds in chemistry [1]: those systems sit at the localised end of exactly this axis, with a Hartree–Fock determinant dominating the ground state.

Geometry is the same knob, not a new one. Comparing a 12-site chain (11 bonds) against a 3×4 grid (17 bonds) at identical sector dimension 924: in the delocalised regime added connectivity strictly hurts (at $\Delta/J = 0.38$ the 2D ground state spreads over 44% of the sector against 1D's 19%, and 2D never reaches the target at all), while in the localised regime ($\Delta/J \geq 2$) the two geometries are *identical* (16.2% and 2.7% in both) — once the ground state collapses onto a handful of configurations, how they are connected no longer matters.

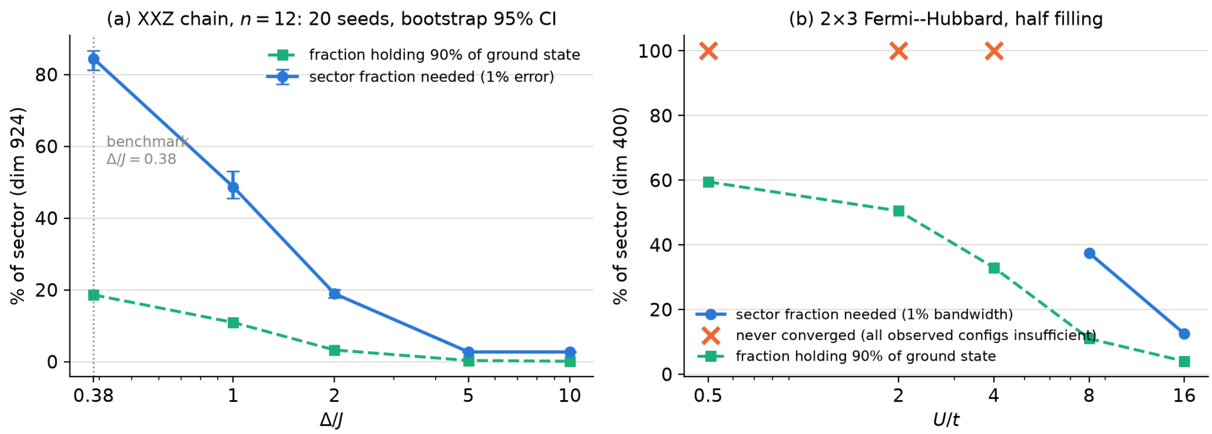

Figure 1: The localisation boundary. (a) XXZ chain, $n = 12$ half-filled sector (dimension 924): sector fraction needed for 1% ground-energy error versus Δ/J, mean over 20 independent sampling seeds with bootstrap 95% confidence intervals, alongside the fraction of the sector holding 90% of the exact ground state. (b) The same boundary on the 2×3 Fermi–Hubbard lattice at half filling (sector dimension 400) as a function of U/t; crosses mark points where no sampled subspace — including the span of *every* configuration the sampler observed — reached the target.

4 Premise and success are separable

A subtlety worth its own section: on the delocalised side, the method's premise *improves* with system size while its performance *degrades*. Across $n = 10, 12, 14$ at $\Delta/J = 0.38$, the fraction of the sector holding 90% of the ground state falls ($24.2\% \to 18.6\% \to 14.1\%$ — concentration, exactly what the method assumes), yet the fraction required for 1% accuracy rises ($79.4\% \to 86.6\% \to 93.2\%$). The gap between those two curves is the ranking quality of the sampler: time-evolved reference states visit the important configurations, but do not visit them *preferentially enough* for a frequency ranking to put them first. Any proposal to push sample-based methods into delocalised regimes must attack the ranking, not the concentration.

Two methodological cautions from our own false starts, recorded because both faked a different conclusion before being caught. First, pooling every configuration a fixed shot budget happens to observe (rather than ranking by frequency and truncating) measures the sampler's coverage, not the method, and made the failure look far worse than it is. Second, on Hubbard, an accuracy threshold relative to $|E_0|$ tightens without bound at large U/t (where $E_0 \sim -t^2/U \to 0$) and reported "never converged" even in the trivially-localised regime; rescaling to the sector bandwidth restored a metric comparable across the axis.

5 The boundary on hardware

We tested the boundary itself on a superconducting processor: one point on each side ($\Delta/J = 0.38$ and 5.0) at $n = 12$, sampling configurations on-device from the Néel reference evolved by second-order Trotter circuits (262 two-qubit gates after routing) to the ten protocol times, 3 000 shots each, with the ideal-Trotter predictions pre-registered before submission (64.9% and 2.7% of the sector, respectively).

Three findings. First, 86% of all shots violate charge conservation and are discarded by sector post-selection — against 1.65% on a 15-gate circuit in an earlier run, so the discard rate tracks depth as expected. Second, *the boundary ordering survives*: the localised point converges (at $600/924 = 64.9\%$ of the sector) while the delocalised point never does. Third, the localised point's requirement is $24\times$ the ideal prediction — and a 20-seed control attributes this cleanly to noise rather than to the reduced sample budget: subsampling the *noiseless* reference protocol to exactly the hardware's kept-in-sector count still converges at a mean of 26 configurations (range $[25, 50]$, $20/20$ seeds). The excess is charge-*conserving* corruption — errors invisible to the symmetry check — polluting the frequency ranking. (At the delocalised point the same subsampled control never converges either, so no noise attribution is claimed there.)

This sharpens the folklore that sample-based methods degrade gracefully under noise: a corrupted sample is *not* always "just an irrelevant basis state." Out-of-sector corruption is detected and discarded for free by conservation; in-sector corruption dilutes the frequency ranking, which is precisely the resource the method depends on — and Sec. 4 shows the ranking is already the binding constraint. Configuration recovery, in other words, removes exactly the errors that matter least. The 2-site configuration-recovery experiment from an earlier run (36 000 shadow shots, 1.65% detected violation rate, recovery validated on a downstream spectral estimate) is consistent with this picture at shallow depth.

6 Work required before submission

We record, deliberately in the manuscript itself at this stage, what this study still needs before the boundary map should be treated as established beyond well-controlled small instances:

1. **Seeds everywhere.** Only the central Δ/J sweep carries 20 seeds and bootstrap intervals; the size scan, geometry comparison, and Hubbard sweep are single-seed and need the same treatment.

2. **Larger sectors.** Dimension 3 432 is far from the regime where sample-based methods claim value; the trend in Sec. 4 should be followed at least another order of magnitude (sparse sector construction makes $n = 16$–20 feasible classically).

3. **Reference-state study.** The Néel-versus-lowest-diagonal comparison was inconclusive at our sizes (the references coincide at $n = 10, 12$); a genuine Hartree–Fock-like family of references with tunable ground-state overlap would turn the chemistry analogy from plausible to demonstrated.

4. **Comparison against production SQD implementations,** including their configuration-recovery and self-consistent iteration machinery, on the same models — our top-M-by-frequency protocol is deliberately minimal.

5. **Direct engagement with the chemistry literature:** at least one small molecular system placed on the same localisation axis, so the boundary connects to the regime where these methods are actually used.

6. **Hardware repeats and error mitigation.** The on-device boundary run (Sec. 5) is a single job on one device; repeats, a second device, and a test of whether dynamical decoupling or readout mitigation reduces the in-sector corruption that drives the 24× inflation.

7 Discussion

The practical output of this study is a placement rule: before applying a sample-based diagonalisation method, estimate where the problem sits on the localisation axis — Δ/J-like anisotropy for spins, U/t for fermions, reference-configuration dominance in general. On the localised side the methods are excellent and geometry is irrelevant; on the delocalised side the required subspace approaches the full sector, worsens with connectivity and with size, and the bottleneck is the sampler's ranking rather than the ground state's concentration. None of this contradicts the reported chemistry successes — it locates them.

All claims here are graded against exact diagonalisation on purpose, and no advantage over classical computation is claimed anywhere.

Data and code availability

All scripts, evidence files, and the results ledger sourcing every number here (row identifiers embedded as comments in the manuscript source) are public at `https://github.com/mnsh0409/Qiskit-Hackathon-2026`.

Acknowledgments

This work was carried out as part of the Qiskit Hackathon 2026 (Topic 5). We acknowledge the use of IBM Quantum services; the views expressed are those of the authors and do not reflect the official policy of IBM or the IBM Quantum team.

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